

(2½ Hours)

[Total Marks: 75]

- N. B.: (1) **All** questions are **compulsory**.
 (2) Make **suitable assumptions** wherever necessary and **state the assumptions** made.
 (3) Answers to the **same question** must be **written together**.
 (4) Numbers to the **right** indicate **marks**.
 (5) Draw **neat labeled diagrams** wherever **necessary**.
 (6) Use of **Non-programmable** calculators is **allowed**.

1. Attempt any three of the following:

15

- Define and explain the Internet of Things
- "Any sufficiently advanced technology is indistinguishable from magic". Discuss.
- Explain calm and ambient technology using example of Live Wire.
- What is manufactured normalcy field? Explain.
- Differentiate between static IP address and Dynamic IP address.
- Define protocol. Explain the following application layer protocols: HTTP, HTTPS, SMTP, FTP

2. Attempt any three of the following:

15

- Discuss the tradeoffs between cost versus ease of prototyping.
- What are the challenges when we move from prototype to mass production? Explain.
- Discuss open source versus closed source hardware and software. State their advantages and disadvantages.
- Explain the following with respect to prototyping embedded devices: Processor Speed, RAM, Networking, USB, Power Consumption and Physical Size and Form Factor.
- How is development done for Arduino? Explain.
- Compare Raspberry Pi and Arduino.

3. Attempt any three of the following:

15

- Explain the non-digital methods of prototyping.
- What are laser cutters? Explain the main features to consider while choosing a laser cutter.
- Explain the different methods used for 3D printing.
- Discuss the different standards that must be considered while implementing APIs.
- Explain POLLING and COMET.
- Write a short note on Message Queuing Telemetry Transport Protocol.

4. Attempt any three of the following:

15

- Discuss the limitations of memory in embedded devices. How is it managed? Explain.
- What are the concerns regarding performance and battery life while writing code for embedded systems?
- Write a short note on Libraries for embedded systems.
- What is a business model? Who is the business for? Explain.
- Explain the following business models: Make Thing Sell Thing, Subscriptions, Customisation.
- Write a short note on venture capital.

[TURN OVER]

5. Attempt any three of the following:**15**

- a. What are the different software options for designing PCB? Explain.
 - b. Explain the steps for manufacturing PCBs.
 - c. What is the importance of Certification for IoT devices? Explain.
 - d. Explain privacy with respect to Internet of Things.
 - e. Discuss the five critical requirements for sensor commons project.
 - f. Write a short note on cautious optimism.
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